Optimizing UAV-Based Data Collection in IoT Networks With Dynamic Service Time and Buffer-Aware Trajectory Planning

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***Abstract*—Unmanned Aerial Vehicles (UAVs) have become vital tools for data collection in Internet of Things (IoT) networks, enabling efficient monitoring and information acquisition across various domains. However, UAV-assisted IoT networks often face significant challenges such as high data loss, latency, and resource inefficiency due to inadequate buffer management and dynamic service time (DST) allocation for IoT nodes. Existing approaches frequently overlook critical factors such as IoT nodes' energy levels, UAVs' energy constraints, and the dynamic data generation rates of IoT devices. To address these challenges, this article introduces a novel strategy for dynamically optimizing UAV trajectories by integrating real-time data from ground nodes and UAV energy levels. Given the DST and buffer constraints (BC) of IoT devices, optimizing UAV trajectories for data collection is a complex problem. We propose an optimization framework that strategically plans UAV trajectories to minimize service time at designated Rendezvous Points (RPs) and bypasses RPs when necessary to reduce the overall trajectory path, taking into account buffer status and dynamic service time adjustments. To efficiently solve this optimization problem, we employ a meta-heuristic technique known as Path Cheapest Arc with Guided Local Search (PCA-GLS) for UAV route planning within predefined time windows. Extensive simulations demonstrate the effectiveness of our proposed solution in optimizing UAV trajectories and improving data collection performance compared to existing algorithms such as BA-UAV, ACO-MS, and NSGA-II.**

***Index Terms*—IoT, buffer-awareness, unmanned aerial vehicles, data gathering, time-windows.**

I. Introduction

OVER the past decade, the Internet of Things (IoT) has seen a growth in its adoption for various applications, such as environmental monitoring, precision agriculture, disaster management, health care, military surveillance, and other industrial automation [1], [2]. These applications and corresponding IoT Devices (IoTDs), such as smart cameras, smart watches, etc., disseminated across vast areas, gather valuable data often critical for decision-making processes. Specifically, data collection is a crucial operation in the IoT, impacting the performance, functionality, and utility of such systems [3], [4]. However, efficient data collection from IoT nodes presents significant challenges due to individual nodes' limited power, computational capacity, and communication range [5]. Yet, the traditional multi-hop data transmission through these links, especially in geographically scattered IoT, can lead to high energy consumption and network latency. An emerging solution to this challenge is to deploy UAVs as mobile data collectors or gateways in these networks [6]. UAV-assisted IoT systems leverage UAVs equipped with smart sensors, wireless transceivers, and antennas to enhance data collection [7], [8]. The mobility of UAVs enables direct data collection from IoT nodes, effectively overcoming the short communication range problem and reducing the energy consumption associated with multi-hop transmissions [9]. UAVs offer rapid deployment, mobility, and data relay capabilities, optimizing trajectory and energy consumption as data collectors [10]. This approach is particularly beneficial in scenarios with ground sensor obstructions, enabling Line-of-Sight communications during data collection. By employing UAVs as data collectors, closer proximity to ground sensors is achieved, facilitating efficient data retrieval and conserving transmission energy, thereby enhancing energy savings and system efficiency [11], [12].

A. Challenges and Motivations

In UAV-assisted IoT networks, optimizing UAV trajectories is essential to minimize data loss and ensure efficient data collection. However, it presents unique challenges, such as optimizing the UAV's flight path while considering UAVs' energy limit, buffer constraints of IoT nodes, and adhering to time windows for serving IoT nodes [13]. Existing approaches often overlook such factors which leads to suboptimal UAV trajectories and inefficient data collection processes. To address these challenges, some researchers focus on optimizing the UAV's flight path for improved coverage [6], while other researchers consider time-window constraints in trajectory planning. Certain research works also account for the buffer status of sensors in the UAV's data collection

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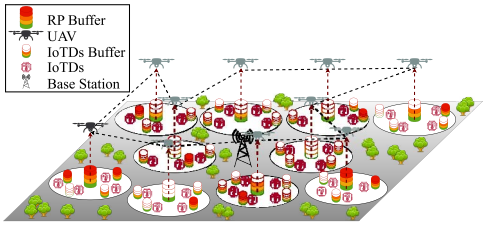

Fig. 1. UAV assisted data collection with RP.

strategy [14]. However, comprehensive strategies that simultaneously address UAV's flight path optimization, time-window constraints, and buffer-aware data collection are still lacking. Another crucial aspect is the selection of RP from IoT nodes, which the UAV visits to collect data, as illustrated in Fig. 1. Our research aims to address these challenges by resolving the buffer-aware data collection problem. It specifically focuses on confronting a pivotal issue in UAV coverage, which involves determining the optimal path for a single UAV to serve multiple RPs within specified time windows.

B. Contributions

The Path Cheapest Arc with Guided Local Search (PCA-GLS) algorithm is leveraged to plan the UAV's trajectory with time windows, enabling IoT nodes to be visited within their respective time windows. Additionally, a buffer-aware strategy is introduced, where the service time of the UAV is adjusted at each RP based on the buffer status of the IoT node. The major contributions to the proposed work are listed as follows:

- Optimal RP Set Selection: The RP selection strategy nominates a minimal set of RPs to cover all the IoT nodes present in the network. The key goal is to optimize the covered path and service time at designated RPs.
- Adaptation of the PCA-GLS: The meta-heuristic method Path Cheapest Arc with Guided Local Search (PCA-GLS) has been employed to efficiently plan the UAV's path considering the time-constrained service at each RP. It will eventually reduce data loss and improve UAV data collection rates.
- Dynamic Service Time Adjustment: The proposed scheme dynamically adjusts the service times of the UAV, further enhancing the efficiency and adaptability of data collection in dispersed IoT networks.

The remaining sections are organized as follows. Section II highlights the state-of-the-art techniques for buffer-aware UAV data collection strategies. Section III defines several parameters and formulates our objective function. Section IV briefly discusses the PCA-GLS method and dynamic service optimization mechanism. The performance analysis of our proposed strategy is illustrated in Section V. Finally, Section VI concludes the paper with a brief discussion on future work.

II. Related Work

Over time, various research efforts on UAV-assisted data gathering in IoT have been conducted, with most studies primarily addressing UAV path-finding challenges, data collection issues, or buffer management problems [15]. To stay within the scope of this paper, two specific categories related to IoT data collection are focused on: (i) UAV-based data collection (ii) Buffer aware data collection.

A. UAV-Based Data Collection

In recent years, many studies have utilized UAVs as mobile data collectors for sensor nodes (SNs), extending network lifetime [16]. Optimization of the wake-up schedule for SNs and the UAV's trajectory aims to minimize the maximum energy consumption of all SNs while ensuring reliable data collection [17]. Despite being cost-effective and flexible, UAVs face challenges with their limited load capacity, particularly for long-term or distant tasks. To overcome this limitation, assembly UAVs are proposed, focusing on optimizing energy resources through system modeling, energy cyclic cooperation, and an in-air energy replenishing strategy [18]. UAV trajectory planning in [6] emphasizes minimizing mission time by optimizing UAV speed, position, and visiting sequence, considering factors like Age of Information (AoI) and on-board energy. Methods such as successive convex approximation (SCA) and genetic algorithms (GA) are used for speed and path optimization, supplemented by an AoI-and-energy-aware trajectory optimization algorithm [13]. Exploration of energy minimization in UAV-assisted sensor networks in [13] involves optimizing trajectory and sensor power while considering constraints using SCA and block coordinate descent (BCD). This includes determining sensor serving orders and UAV hovering positions, with the latter affecting sensor power. Optimal sensor serving orders are determined using the efficient Cutting-Plane method for the standard traveling salesman problem (TSP), while UAV position and sensor power optimization employ the pattern search method for transmit power and SCA for UAV hovering positions. In [19], a very interesting solution for UAV energy constraints was introduced, suggesting a UAV data collector supported by a mobile charging station (UGV) to minimize IoT device AoI using multi-agent deep Q-network (MADQN) to optimizes UAV charging, IoT servicing, and UGV trajectory, considering obstacles and IoT movements. Strategically placing refueling vehicles on the ground provides a promising means to extend UAV operational range, enabling quick recharging without returning to a fixed base.

B. Buffer Aware Data Collection

Buffer management in UAV-assisted IoT networks is crucial for handling transmission variability, preventing data loss, and optimizing data transmission efficiency. It involves temporarily storing data packets during poor transmission conditions to ensure reliable data collection and minimize delays and packet loss. The study in [20] introduces a UAV path planning algorithm for reliable data collection, adjusting trajectories based on the buffer sizes of nodes connected to Access Points (AP) while conserving energy of the nodes, selecting optimal APs, and preventing data loss. It proposes a modified ACO for trajectory design and machine learning for data loss prevention. In the

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context of UAV-assisted edge caching in IoTs, it is emphasized that the simultaneous design of UAV trajectories and caching strategies is essential for improving quality of experience and minimizing content update expenses. A collaborative approach is suggested for this in [21], considering both user experience and content update costs which enables UAVs to efficiently serve areas with high demand for popular content. A queue-aware cooperatively caching mechanism in a multi-UAV-assisted system, addressing random user requests through a well-designed connected dominating set (CDS) for collaborative caching scheduling is introduced in [22]. The proposed work in [23] introduces a trajectory planning algorithm for an energy-efficient Mobile Sink (MS) tour. It considers limited buffer size of the nodes and uniform sensing rates, utilizing virtual grid cell construction to adjust communication ranges based on data buffering. The aim is to minimize the MS traveling path length and solve the nominal MS sojourn scheduling problem within a specified computation time. In [24], sensor nodes (SNs) are managed by cluster heads (CHs) for data collection with UAV support, focusing on minimizing deployment costs and energy usage while meeting communication, UAV mission time, and battery constraints through offline and online UAV-based data collection methods. The study in [14] investigates a mobile relaying system assisted by a UAV with a finite buffer, optimizing the UAV relay node's trajectory to maximize data throughput while accommodating delay-limited and delay-tolerant transmission requirements using SCA and Quadratically constrained programming (QCP). In [9], UAVs with virtual queues address transmission challenges in UAV-assisted IoT networks. A Deep Reinforcement Learning (DRL) model minimizes the Expected Weighted Sum Age of Information for IoT devices by adjusting UAV altitude based on real-time channel conditions.

Existing research primarily addresses individual aspects of data collection, path selection, delay optimization, buffer management and dynamic service time adjustment. Although some studies acknowledge the significance of buffer-aware UAV data gathering in IoT, there is a need for further investigation into challenges such as buffer management with dynamic service time within UAV-based data collection in IoT. These gaps are the focus of our proposed work that integrates an efficient UAV path planning strategy, a buffer-aware scheduling mechanism, and time window considerations. Table I presents a summary of the existing literature regarding UAV-assisted networks, delineating the aims of previous research alongside our proposed approach.

TABLE I
SUMMARY OF THE RELATED WORK AND PROPOSED WORK

Ref.	Objective	Method	Improvement	BC	DST
[19]	AoI	MADQN	Reduced AoI, Energy of UAV and UGV	No	No
[18]	Working time of UAVs & Data Age	RL	Reduced working time of UAVs	No	No
[17]	Energy consumption of UAV & nodes	SCA	State aware trajectory planning	No	Yes
[6]	AoI & Energy aware trajectory	SCA & GA	Reduced mission time and energy consumption	No	No
[20]	Buffer aware UAV path planning	ACO & ML	Reduced energy consumption,data loss & latency	Yes	Yes
[9]	Expected Weighted Sum AoI	DRL	Optimized altitude of UAV while minimizing the AoI	Yes	No
[21]	Cache aware path planning of UAVs	MA-DRL	Improved QoE	Yes	No
Our Work	Path optimization based upon DST & Buffer	PCA-GLS	Buffer aware & service time based path optimization	Yes	Yes

TABLE II
DEFINITIONS OF COMMON NOTATIONS

Symbol	Definition
v_i	A vertex in the graph G, representing an RP
e_i	Earliest arrival time at RP v_i
l_i	Latest arrival time at RP v_i
B_i	Buffer capacity at RP v_i
b_i	Buffer status at RP v_i
T_f	Service Time when Buffer is Full
T_p	Service Time when Buffer is not Full and chord is formed
T_n	Fixed Service Time when UAV is not passing through the interior of RP's communication range
D_{mat}	Distance Matrix
d_{ij}	Travel time between locations i and j

III. SYSTEM MODEL AND PROBLEM FORMULATION

In the proposed scenario, we consider an environment where a total of N stationary IoT nodes are strategically dispersed across the designated area for monitoring. This configuration includes the deployment of one UAV and a centralized depot. This optimized setup ensures comprehensive coverage of the monitoring zone, guaranteeing robust and reliable data collection. In the proposed work, we have considered a hybrid mode for UAV data collection, which includes flying mode and hovering mode [3].

A. Symbols and Definitions

The subset of IoT nodes that are selected as RPs as a graph is denoted as $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_n\}$ is the set of vertices (RPs), and $E = \{e_1, e_2, \ldots, e_m\}$ is the set of edges representing the travel time between RPs. Each vertex v_i has a corresponding time window $[e_i, l_i]$ (earliest and latest arrival times), a buffer capacity B_i, and a buffer status b_i. The UAV starts and ends its route at a designated depot node, denoted as v_0. The aim is to find a route sequence $R = \{v_0, v_1, \ldots, v_n, v_0\}$ that minimizes the total travel distance, respects the time window constraints, and optimizes service time based on buffer status. Table II illustrates a short list of related notations and their meanings.

B. Energy Consumption Model

In the IoT network, overall energy efficiency is a critical performance metric due to the limited energy resources of the nodes and potentially the data-collecting entities like UAVs. In the proposed system, the IoT nodes and the UAV incur significant energy costs, which impact the system's operational performance and lifespan. Therefore, it is essential to accurately model the energy consumption of these key components and devise strategies to optimize them. For the

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IoT nodes, we adopt the first-order radio model, a widely used energy consumption model in IoT proposed in [20].

The energy usage of UAV is affected by traveling to Rendezvous Points (RPs) and transferring data to IoTDs. The primary energy depletion in UAVs results from flight mechanics and data transmission. In our model, we adopt a mechanical energy consumption model based on propulsion as in [25] which considers the energy consumption dependent on acceleration for rotary-wing UAV. The mechanical energy consumed by the UAV over N time slots is represented as E_{ME}.

$$E_{ME} = \Delta \sum_{i=1}^{N} wc\|a[i] - a_g\|^2 \tag{1}$$

where a represents the acceleration vector and a_g denotes the vector due to gravity acceleration and Δ is the length of the time slot. w denotes the energy consumption coefficient attributed to resistance induced by factors like wind, rain or other weather conditions. $w = 1$ signifies ideal weather conditions; otherwise, $w > 1$. c is the energy consumption coefficient associated with UAV properties such as mass, moment of inertia, etc. Thus the total energy consumed by the UAV in N time slot is given by following equation.

$$UAV_E = \Delta \sum_{i=1}^{N} wc\|a[i] - a_g\|^2 + P_{tx} N \Delta \tag{2}$$

where P_{tx} represents the energy consumption due to the data transmission.

C. Communication Model

In the context of UAV-based data collection from selected RPs, we need to establish a communication model between the UAV and the IoT Nodes present in the network [26]. The signal transmission channel is influenced by factors such as line-of-sight (LoS) and non-line-of-sight (NLoS) links, and the path loss is deduced as follows.

$$L_\gamma\big(d_{u,n}(t)\big) = \begin{cases} \left(\frac{4\pi f d_{u,n}(t)}{c}\right)^2 \delta_0, \ \gamma = 0 \\ \left(\frac{4\pi f d_{u,n}(t)}{c}\right)^2 \delta_1, \ \gamma = 1 \end{cases} \tag{3}$$

Here, f represents the carrier frequency, c denotes the speed of light, and δ_0 and δ_1 represent the NLoS and LoS links' excessive path losses. The coordinates can be used to determine the distance $d_u, n(t)$ between the UAV hovering over RP c_n and the scheduled IoT node I_u:

$$d_{u,n}(t) = \sqrt{(x_u(t) - x_n(t))^2 + (y_u(t) - y_n(t))^2 + h^2} \tag{4}$$

The data uploading process begins once the UAV reaches the RP c_n and establishes a communication connection with IoT node I_u. P_t denotes the transmission power of IoT node I_u. The data uploading rate for s_u is expressed as following.

$$R_{u,n}(t) = B \log_2\left(1 + \frac{\alpha P_t}{L_\gamma\big(d_{u,n}(t)\big)\sigma^2}\right) \tag{5}$$

Here, α represents the channel power gain at the reference distance of 1 meter, and B and σ^2 are the system bandwidth and the noise power at the UAV receiver, respectively. To further analyze the efficiency of data uploading time, we introduce an equation as the ratio of the data buffered data size to the data uploading rate, denoted as $D_{u,n}(t)$.

$$D_{u,n}(t) = \frac{B_i}{R_{u,n}(t)} \tag{6}$$

Here, B_i represents the amount of buffered data that is uploaded from IoT node I_u to the UAV. By considering this ratio, we can evaluate the effectiveness of the data uploading process in terms of the allocated transmission time.

D. Channel Model

In our model we have used Rician channel fading model (RCF) which is crucial in UAV scenarios, capturing both LoS and scattered signal components. While UAV communication resembles terrestrial channels, factors like mobility and altitude influence its characteristics. At higher altitudes, clear LOS communication prevails, resulting in a higher Rician K-factor. Nevertheless, multipath propagation persists, causing fading effects. Altitude changes can intensify these effects. As UAVs maneuver, channel conditions fluctuate rapidly, demanding adaptable communication systems. The channel fading experienced by IoT devices and UAVs comprises both large-scale and small-scale fading components defined as following equation [27].

$$F = \sqrt{(P_L \times S_F)}.h \tag{7}$$

where P_L refers to signal strength attenuation, considering factors such as observation height, angle, propagation distance and fuselage structure, while variable S_F accounts for shadowing caused by obstacles along the propagation path and h represents the channel impulse response between the transmitter and receiving antennas, expressed as the combination of both LoS and NLoS components in the following equation as in Rician channel fading model.

$$h = \sqrt{\frac{K}{K+1}} \cdot h_L + \sqrt{\frac{1}{K+1}} \cdot h_{NL} \tag{8}$$

where, h_L denotes the complex channel gain for the LoS component, h_{NL} represents that for the NLoS or scattered components and K is the Rician K-factor which determines the severity of fading, with higher values indicating stronger line-of-sight dominance and lower values suggesting increased scattering.

E. Problem Formulation

The key objective of this work is to optimize the travel distance while minimizing service delay and buffer data for UAVs. The objective function is formulated as follows:

1) *Input:*
   - A set of locations $N = \{0, 1, \ldots, n+1\}$, where location 0 and location $n+1$ are the depot, and the other locations represent the visiting nodes.
   - A distance matrix $D_{\text{mat}} = [d_{ij}]_{n+1 \times n+1}$ representing the travel time or cost between locations i and j. It is typically assumed that $d_{ij} = d_{ji}$ for all $i, j \in N$, and $d_{ii} = 0$ for all $i \in N$.

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- A set of time windows $W = \{[e_i, l_i]\}_{i=1}^{n}$, where e_i and l_i represent the earliest and latest time that service can begin at location *i* respectively.
- A set of service times $S = \{s_i\}_{i=1}^{n}$, where s_i is the time required to serve node *i*.

2) *Decision Variables:*
   - $x_{ij} = 1$ if the UAV travels from location *i* to location *j*, and 0 otherwise.
3) *Objective:*
   Minimize $\sum_{i,j \in N} d_{ij} x_{ij}$, which is the total travel time or cost for UAV networks.
4) *Constraints:*
   - Each node is visited exactly once: $\sum_{i \in N} x_{ij} = 1$ for all $j \in N \setminus \{0, n+1\}$.
   - The UAV starts and ends its route at the depot: $\sum_{j \in N} x_{0j} = \sum_{i \in N} x_{i(n+1)} = 1$.
   - Time window constraints are: $e_j \leq t_j \leq l_j - s_j$ for all $j \in N \setminus \{0, n+1\}$.
   - The data uploading time $D_{u,n}(t)$ must be bounded by maximum tolerable delay.
5) *Buffer Management* The service time (S_i) at each node *i* depends on the buffer status (b_i) and is defined as:

$$S_i = \begin{cases} T_f, \text{ if } b_i = B_i \\ T_p, \text{ if } b_i < B_i \\ T_n, \text{ if } b_i < B_i \end{cases} \tag{9}$$

IV. Proposed Methodology

From the objective function, we can easily observe that finding the shortest possible path among the RPs is complex and mostly NP-hard. Due to their computational difficulty, no effective algorithms can solve NP-hard problems in polynomial time. Finding optimal solutions for large instances of NP-hard problems is computationally impossible, but several algorithmic techniques exist to get close to the optimal solution. Therefore, we propose a UAV-Based Data collection scheme with Dynamic Service Time and Buffer-Aware trajectory planning (DST-BA). In the proposed solution, we have introduced heuristic algorithms: (A) Selecting RPs from a given set of nodes, (B) Planning the UAV's Path considering Time Windows, and (C) Dynamic Service Time Adjustment Algorithm. The key objective is to obtain a near-optimal solution in complex UAV communication networks.

A. Selection of RPs

In the proposed approach to route planning, a key consideration is the selection of RPs, as illustrated in Fig. 2. These points are predefined locations in the network of nodes that serve as gathering or checkpoint locations for the UAV during its mission. The key objective here is to select a minimal set of RPs that covers all IoT nodes in the network. The Algorithm 1 represents the procedure for the proposed RP selection in this work. The Euclidean distance d_{ij} between nodes *i* and *j* is calculated as follows.

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{10}$$

where (x_i, y_i) and (x_j, y_j) represent the coordinates. Algorithm 1 calculates the neighborhood for each node based on the communication radius. It then iteratively selects the node with the highest number of neighbors as an RP and removes this node and its neighbors from consideration as RPs. This process continues until all nodes are covered by the selected set of RPs, forming the set of all RPs. The selection of suitable RPs for the UAV is a crucial step in effective data collection from the IoT nodes.

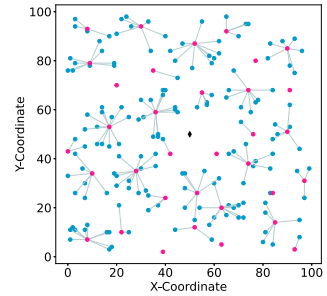

Fig. 2. Illustration of RP selection over the network.

Algorithm 1 Rendezvous Point Selection

Require: Nodes $\mathcal{N}$, Location (x, y), Maximum radius R_{max}
Ensure: A set of Rendezvous Points *RP*

```
Initialize an empty set RP.
for each node n_i in N do
    Calculate its set of neighbors N_{n_i},
end for
while N is not empty do
    Select the node n* in N with the largest N_{n*}.
    Add n* to RP.
    Remove n* and all nodes in N_{n*} from N.
    for each remaining node n_i in N do
        Update N_{n_i} to remove nodes that are not in N.
    end for
end while
return RP
```

- *Reduction of Redundant Data:* By collecting data only from IoT nodes within R_{max}, the UAV avoids redundant data collection from distant nodes. This helps to reduce noisy data.

B. UAV Path Planning With Time-Windows

The UAV path planning problem with time windows involves finding the optimal route for a single UAV to serve a number of nodes within specified time windows while minimizing the total travel cost. This can be achieved using a method called Path Cheapest Arc with Guided Local Search.

- *Route Construction with Path Cheapest Arc:*

Starting from the current node *i* (initially the depot), the algorithm identifies the node *j* that is not yet included in the route (*R*) and has the lowest travel cost (referred as "cheapest arc"). It ensures that visiting this node does not violate the time window constraints. The algorithm then adds node *j* to the route and proceeds to the next iteration from the newly added node. This process continues until all nodes have been visited. This strategy forms the essence of the Path Cheapest Arc algorithm.

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Path Cheapest Arc

- Construct a route R.
- Start at the depot (node 0).
- While there are unvisited nodes that can be serviced without violating the time window constraints:
  - Find the node j not in R that minimizes $D[i][j]$ (the cost of travel from the current node i), subject to the constraints $arrival_time + D[i][j] + s_j \leq l_j$.
  - Add node j to R, update the UAV's arrival time at the next node, and mark node j as visited.
- When no more nodes can be added, return to the depot.

- *Solution Improvement with Guided Local Search:* In the GLS algorithm, penalties are added to certain features of the solution in order to guide the search towards unexplored areas of the solution space and help it escape from local minima. The use of high penalties can discourage the selection of certain routes or nodes in the solution. When dealing with the UAV routing problem with time windows, one important aspect to consider is the tardiness at each node, which refers to the amount of arrival time exceeding the node's deadline. If a solution involves arriving late at a node, it can incur a penalty. Initially, the algorithm may prioritize routes that result in some late arrivals, as it allows for servicing more nodes and reducing the total travel distance. However, when a high penalty is applied to tardiness, the cost of such solutions increases. This prompts the algorithm to reject these solutions during the local search process and instead select alternatives that meet all deadlines, even if they require longer travel distances. This approach strikes a balance between minimizing travel distance and meeting deadlines, leading to more practical and reliable solutions for real-world applications. In our proposed approach, we have incorporated a penalty for violating time window constraints at each location.

Guided Local Search

- Given the initial solution R, apply the Guided Local Search (GLS) metaheuristic.
- Perform a local search operation on R to find a neighboring solution R' that improves the total cost.
- If no improving neighbor is found, assign penalties to certain features of the current solution.
- Continue the local search with the penalized objective function $f + \lambda p$, where λ is a penalty factor. Repeat this process until a stopping criterion is met, such as a maximum number of iterations or a time limit.

The final output of GLS is the best solution found during the search, R^*, which is a route for the UAV that minimizes the total cost, including penalties.

Algorithm 2 PCA-GLS

Require: N: nodes, D: distance matrix
Ensure: R^*: optimal route

```
R ← empty route
i ← 0 {start at depot}
while there are unvisited nodes do
    Path Cheapest Arc()
end while
Return to depot
Apply GUIDED_LOCAL_SEARCH to R, obtain R'
while improving neighbor exists in R' do
    Find improving neighbor, update R'
end while
Penalize features if no improving neighbor
Repeat search with penalized objective function until stopping criterion met
R* ← best solution found during search
return R*
```

It is important to note that this penalty does not directly modify the fitness function of the GLS algorithm. The detailed steps of the UAV path planning while considering time windows are presented in Algorithm 2.

C. Dynamic Service Time Adjustment

The proposed research builds upon and enhances this concept by introducing dynamic service time adjustment. Traditional approaches often assume a fixed service time for data gathering, irrespective of the data volume or buffer status of the node. In contrast, the proposed strategy dynamically adjusts the service time based on the buffer status and the distance from the depot to the next point. By choosing to navigate to the center of the node's communication range when the buffer is full and to the nearest point on the communication range when the buffer is not full, the proposed strategy has the potential to save more time and power.

The nearest point on the edge of a circle to a given point is calculated using vector normalization. Given the center of the circle (c_x, c_y), the radius R_c, and the given point (p_x, p_y), the vector from the center to the point is $\vec{v} = (p_x - c_x, p_y - c_y)$. The distance d between the center and the point is then calculated using the Pythagorean theorem.

$$d = \sqrt{(p_x - c_x)^2 + (p_y - c_y)^2}. \tag{11}$$

The normalized vector is then $\hat{v} = \frac{1}{d}\vec{v} = (\frac{p_x - c_x}{d}, \frac{p_y - c_y}{d})$. The nearest point (n_x, n_y) on the edge of the circle to the given point is then R times the normalized vector away from the center.

$$n_x = c_x + R_c \cdot \frac{p_x - c_x}{d}, \tag{12}$$

$$n_y = c_y + R_c \cdot \frac{p_y - c_y}{d}. \tag{13}$$

This strategy aims to dynamically adjust the service times at RPs based on their buffer status and relative positions, as illustrated in Fig. 3. Algorithm 3 presents the internal steps of the dynamic service time adjustment strategy. The inputs

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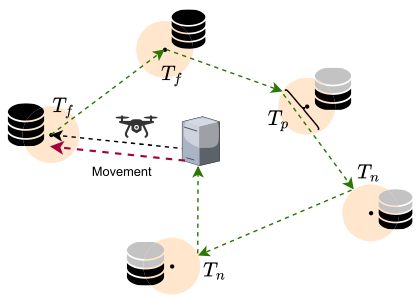

Fig. 3. Buffer-aware path and adjusted service time.

Algorithm 3 Dynamic Service Time Adjustment

Require: Sets: Coordinates of RPs C, Visiting sequence S, Buffer status list B, Initial service times T
Require: Starting point D, Radius R_c, UAV velocity V
Initialize service times list $ST = T$
for $i = 1$ to n **do**
  $C_{\text{current}} = S_i$
  if $B_i == 1$ **then**
    $C_{\text{next}} = C_{\text{current}}$
    $L_{\text{extra}} = \frac{2 \times R_c}{V}$ {Time for extra travel}
    if $T_i \leq L_{\text{extra}}$ **then**
      $ST_i = 0$ {Adjust service time}
    else
      $ST_i = T_i - L_{\text{extra}}$ {Adjust service time}
    end if
  else
    $C_{\text{next}} = \text{find_nearest_point}(C_{\text{current}}, R_c, D)$
    if path from D to C_{next} forms a chord in the circle with center at C_{current} **then**
      $L_{\text{chord}} = \text{get_distance}(D, C_{\text{next}})$
      if $T_i \leq \frac{L_{\text{chord}}}{V}$ **then**
        $ST_i = 0$ {Adjust service time}
      else
        $ST_i = T_i - \frac{L_{\text{chord}}}{V}$ {Adjust service time}
      end if
    end if
  end if
  $D = C_{\text{next}}$
end for
return ST {Return adjusted service times}

to the algorithm include the coordinates of RPs, the visiting sequence, the buffer status list, the depot as the starting point, initial service times, the radius of circles, and the UAV velocity.

Set $C = C_1, C_2, \ldots, C_n$ denote the coordinates of RPs, $S = S_1, S_2, \ldots, S_n$ as the visiting sequence, $B = B_1, B_2, \ldots, B_n$ as the buffer status list, where $B_i = 1$ if the buffer at C_i is full and $B_i = 0$ otherwise, D as the depot, $T = T_1, T_2, \ldots, T_n$ as the initial service times, R_c as the radius of the circles, and V as the UAV velocity. The algorithm initially sets the service times list as $ST = T$. For each C_i in the visiting sequence, the algorithm considers two main cases:

1) If the buffer at C_i is full ($B_i = 1$), the UAV has to go to the center of the circle, offload the data, and then continue to its next destination. This deviation to the center and forward again adds an extra distance of $2R_c$, resulting in an extra travel time of $\frac{2R_c}{V}$. If the initial service time T_i is less than or equal to this extra time, the service time at C_i is adjusted to 0 ($ST_i = 0$). Otherwise, it is decreased by the extra time ($ST_i = T_i - \frac{2R_c}{V}$).
2) If the buffer at C_i is not full ($B_i = 0$), the UAV can fly directly to the edge of the next rendezvous point without going to the center. If this flight forms a chord in the circle with the center at C_i, the length of the chord L_{chord} is calculated. If the initial service time T_i is less than or equal to the time required to travel along the chord $\frac{L_{\text{chord}}}{V}$, the service time at C_i is adjusted to 0 ($ST_i = 0$). Otherwise, it is decreased by the chord travel time ($ST_i = T_i - \frac{L_{\text{chord}}}{V}$). If a chord is not formed, the UAV serves the service time at the entry point of the circle. After visiting all RPs in the sequence, the algorithm returns the adjusted service times ST.

We must determine the exit point to calculate the chord length within the current circle by finding the point on the circle's boundary closest to the next destination. $P_{\text{current}} = (x_c, y_c)$, $C_{\text{next}} = (x_n, y_n)$, and the exit point from the current circle is $P_{\text{exit}} = (x_e, y_e)$. Further, x_e and y_e is defined as.

$$x_e = x_c + R_c \times \frac{x_n - x_c}{d}, \tag{14}$$

$$y_e = y_c + R_c \times \frac{y_n - y_c}{d}, \tag{15}$$

where $d = \sqrt{(x_n - x_c)^2 + (y_n - y_c)^2}$ and R_c is the circle's radius. The chord length L_{chord} is calculated as follows.

$$L_{\text{chord}} = \sqrt{(x_e - x_c)^2 + (y_e - y_c)^2}. \tag{16}$$

If the UAV is currently at an RP with an empty buffer, and the next RP is located in a way that a chord cannot be formed within the current RP circle, then the UAV should head directly to the next RP.

Lemma 1 (Constrained UAV Path): $R = r_1, r_2, \ldots, r_n$ represent a set of RPs, and $W = w_1, w_2, \ldots, w_n$ represent a set of time windows corresponding to each point, where each $w_i = [s_i, e_i]$ represents the start and end times of the window. For any path P of the UAV, if the travel time to RP r_i ($t_{\text{travel}i}$) plus the time spent at the previous RP ($T\text{prev}$) exceeds the end time of the time window (e_i), then r_i can be skipped for that round, *i.e.,* not included in path P.

Proof: Assume a path P for the UAV such that for each RP $r_i \in P$, the UAV arrives at r_i within its corresponding time window w_i. The arrival time at r_i is given by:

$$t_{\text{arrival}_i} = T_{\text{prev}} + t_{\text{travel}_i} \tag{17}$$

We assume that for all $r_i \in P$, $t_{\text{arrival}_i} \leq e_i$. Now consider an RP r_j for which

$$T_{\text{prev}} + t_{\text{travel}_j} > e_j \tag{18}$$

According to the assumption, r_j is expected to be included in the path P. However, if the UAV cannot reach r_j within the time window w_j, it contradicts the requirements of path P.

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Therefore, for such r_j, it can be excluded from path *P* for that round. This leads to the conclusion that under time window constraints, the UAV path may involve skipping certain RPs that cannot be feasibly visited within their time windows. ■

Lemma 2 (Optimal RP Selection): Given a set of RPs $R = r_1, r_2, \ldots, r_n$, and a cost function $C(r_i, r_j)$ that represents the cost of traveling from point r_i to point r_j, the optimal choice for the next RP r_k from the current position r_i is determined using the following formula:

$$r_k = \arg \min_{r_j \in R, r_j \neq r_i} C(r_i, r_j) \tag{19}$$

subject to time window constraints. This means that the UAV aims to minimize the travel cost while ensuring compliance with the specified time constraints.

Proof: We will prove this lemma through contradiction. Let's assume the existence of a RP r_m such that $C(r_i, r_m) < C(r_i, r_k)$, but the UAV does not select r_m. If r_m satisfies the time window constraints, choosing r_m instead of r_k would result in a lower overall cost, which contradicts our assumption that the UAV minimizes travel cost. Hence, if there is a point r_m with a lower cost, it must not satisfy the time window constraints. Therefore, the UAV always selects the point with the lowest cost while adhering to the time window constraints. ■

Theorem 1 (Feasible Path Under Time Window Constraints): Given a set of RPs $R = r_1, r_2, \ldots, r_n$ and a cost function $C(r_i, r_j)$, a feasible path exists for the UAV that satisfies time window constraints if the following condition holds:

$$\sum_{i=1}^{n-1} C(r_i, r_{i+1}) + C(r_n, r_1) \leq T \tag{20}$$

where *T* represents the total allowable travel time for the UAV, the existence of such a path is guaranteed by the Optimal RP Selection lemma, provided that the time window constraints are met at each RP.

Proof: For a path to be feasible, the total time taken to travel the path must be less than or equal to *T*. We can write this condition as:

For each pair of points (r_i, r_{i+1}), according to the lemma, we choose r_{i+1} such that $C(r_i, r_{i+1})$ is minimized while satisfying the time window constraints. Let us denote this minimum cost as $C_{\min}(r_i)$. Therefore, if the following condition holds:

$$\sum_{i=1}^{n-1} C_{min}(r_i) + C_{min}(r_n) \leq T \tag{21}$$

From Eqn. (21), it can be observed that there exists a feasible path, which concludes our proof. ■

V. Discussion of Complexity

In this section, we discuss the computational complexity of the proposed strategy in terms of path planning and time adjustment in UAV-enabled data collection.

1) Path Planning Complexity: The path planning phase involves the selection of RPs and finding the optimal route for the UAV to serve the nodes within specified time windows. The selection of RPs is based on calculating neighborhoods and choosing nodes with the highest number of neighbors as RPs. This step has a complexity of $O(|\mathcal{N}|)$, where $|\mathcal{N}|$ is the number of nodes in the network. The subsequent route construction using the Path Cheapest Arc algorithm has a complexity of $O(|\mathcal{N}|^2)$, taking into account the calculation of distances between all pairs of nodes. The complexity of the GLS algorithm for solution improvement depends on the specific implementation, typically ranging from $O(|\mathcal{N}|^2)$ to $O(|\mathcal{N}|^3)$. Therefore, the overall complexity of the path planning phase can be expressed as $O(|\mathcal{N}|^3)$.

2) Time Adjustment Complexity: The dynamic service time adjustment algorithm considers the buffer status and distance from the depot to determine the service times at RPs. This algorithm involves iterating through all RPs, calculating distances, and adjusting service times accordingly. The complexity of this algorithm is $O(|RP|)$, where $|RP|$ is the number of RPs.

3) Overall Complexity: The proposed methodology comprises the path planning and time adjustment phases. The complexity of the path planning phase dominates the overall complexity. Therefore, the overall complexity of the proposed algorithm can be approximated as $O(|\mathcal{N}|^3)$.

VI. Performance Evaluation

This section presents the simulation results of our proposed UAV-Based Data collection method with Dynamic Service Time and Buffer-Aware trajectory planning (DST-BA). To analyze the performance of the DST-BA, we have considered several performance matrices, such as network lifetime, data collection rate, and path length. Further, to present the effectiveness, we have compared DST-BA with current research works such as BA-UAV [28], ACO-MS [5] and NSGA-II [20]. This comparative review allowed us to position the proposed research within the existing academic landscape and evaluate the relative strengths of the proposed scheme.

A. Experimental Setup

The proposed work used extensive simulation exercises powered by Python 3.9.13. The computational environment in which these simulations were carried out included an operating system of Windows 10 Enterprise 64-bit, a memory capacity of 8 GB RAM, and an Intel Core i5 processor. In the experiment, N stationary IoT nodes were randomly distributed in a region of 100 × 100 m^2. The base station was strategically fixed at the coordinates (50,50) m, serving as a constant reference point. The principal methodology of this study entailed the application of a meta-heuristic approach to determine an initial UAV flight path. Subsequently, this path was optimized by implementing a Dynamic Service Time Adjustment algorithm, further refining our outcomes. Simulation parameters that are used in our experiment are shown in Table III [20].

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TABLE III
SIMULATION PARAMETERS

Parameter	Value
Velocity of the UAV for horizontal motion (V_h)	2 m/sec
Altitude of UAV's hovering (U_h)	10 m
Velocity of the UAV for vertical motion (V_v)	1 m/sec
Size of the Data packet	5000 bits
Size of the Control message	200 bits
Node energy (E_o)	0.5 J
Filed Dimensions	100×100 m^2
Number of nodes (N)	200,400,600 nodes
Path Loss Reference distance (D_o)	15 m

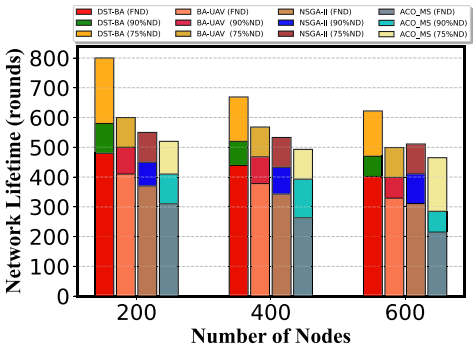

Fig. 4. Analysis of network lifetime for different nodes.

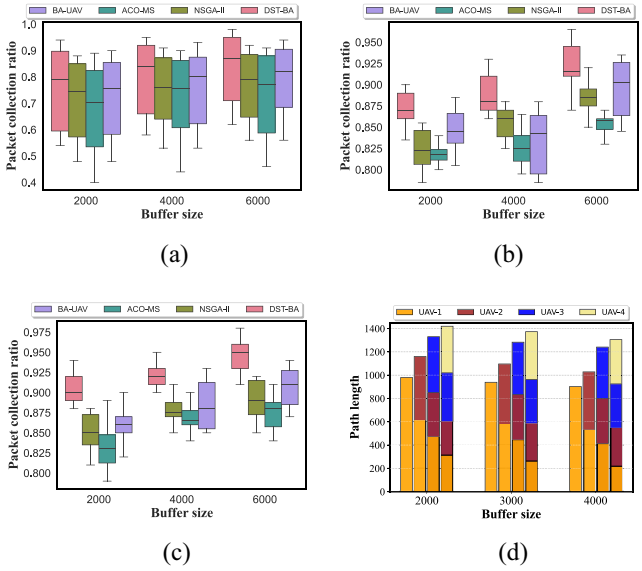

Fig. 5. Performance analysis of data collection rate when (a) Number of nodes is 200. (b) Number of nodes is 400. (c) Number of nodes is 600, and (d) Path length by varying number of UAVs.

B. Network Life Time

Since IoT devices have limited energy and transmission range, UAVs are used to collect data by hovering at IoT nodes. This helps to conserve transmission energy across all IoT nodes. In the event of a buffer overflow, all neighboring nodes associated with the RP will be required to re-transmit their data to their respective RP. This process significantly increases energy consumption, thus reducing the lifetime of the network. Fig. 4 graphically represents how network lifetime is impacted by an increase in the number of nodes. As the number of nodes grows, there is a subsequent increase in neighboring nodes for several RPs, which consequently leads to a higher incidence of buffer overflows, adversely affecting the network lifetime. The DST-BA integrates time windows and buffer awareness for data gathering, a strategy that significantly mitigates the prevalence of buffer overflows. Compared to methodologies presented in [28] and [5], the DST-BA demonstrates lower buffer overflow rates, which contributes to better energy preservation, resulting an extension of the network lifetime.

C. Data Collection Rate

The proposed methodology handles buffer overflows efficiently by integrating time windows and buffer awareness, resulting in a better data collection rate. When the buffer size is increased, it prevents data loss due to buffer overflow, which helps in improving the data collection rate. If data generation rates at the nodes exceed the UAV's collection rate, increasing the buffer size can prevent data loss due to buffer overflow. When the buffer is full, incoming data can be lost until there is enough room again in the buffer. Fig. 5(a) represents the data collection rate for different buffer sizes when the number of nodes is 200. The DST-BA gave better results for the Data collection rate. Moreover, the potential data loss is reduced as the buffer size increases. If UAV is interrupted or unable to provide service at a particular RP, large buffers continue to store the data avoiding buffer overflows resulting in a better data collection rate. Fig. 5(b) and Fig. 5(c) represent the data collection rate for different buffer sizes when the number of nodes is 400 and 600, respectively. With larger buffers, nodes can store more data between UAV visits. As a result, during each visit, the UAV can collect more data, increasing the effective data collection rate.

D. Effect of Path Length

Fig. 5(d) represents the trajectory length for varying UAVs used 1, 2, 3, and 4 respectively for different buffer capacity. It should be noted that the overall travel length grows when more UAVs are employed. It is now owing to an increase of overlapping flying regions closer to the depot. As a result, adding unneeded UAVs for data gathering will not only add to the cost of deployment but will also increase the overall energy consumption of the UAV system. When the buffer size is increased, there is a high chance of optimizing the path length. With more data stored at each node, there is more flexibility in scheduling the UAV's path. This means that the UAV can be scheduled to visit nodes in a way that minimizes the total path length, rather than needing to visit certain nodes urgently to prevent buffer overflow.

E. Data Generation & Buffer-Aware Path

In Fig. 6(a), Fig. 6(e), and Fig. 6(i), the red color represents a higher data generation rate. These three are taken from a random UAV round number to visualize how the data generation rate changes. Fig. 6(b), Fig. 6(f) and Fig. 6(j) represent RP generation, Fig. 6(c), Fig. 6(g) and Fig. 6(k) represent the corresponding path formed while considering time-windows, Fig. 6(d), Fig. 6(h) and Fig. 6(l) represents how the obtained

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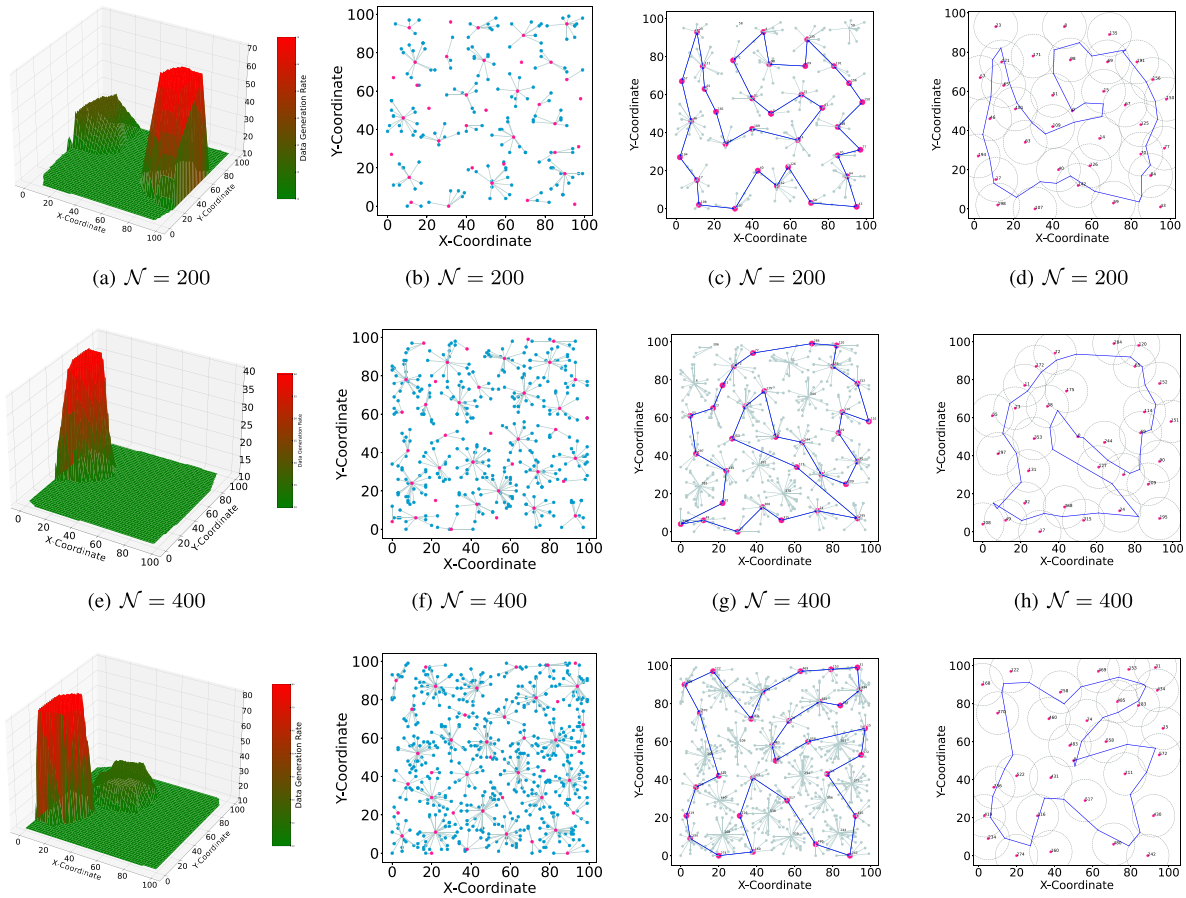

(a) $\mathcal{N} = 200$ (b) $\mathcal{N} = 200$ (c) $\mathcal{N} = 200$ (d) $\mathcal{N} = 200$

(e) $\mathcal{N} = 400$ (f) $\mathcal{N} = 400$ (g) $\mathcal{N} = 400$ (h) $\mathcal{N} = 400$

(i) $\mathcal{N} = 600$ (j) $\mathcal{N} = 600$ (k) $\mathcal{N} = 600$ (l) $\mathcal{N} = 600$

Fig. 6. Performance analysis (a) Heat-map for data generation rate. (b) RP generation. (c) Path formation. (d) Adjusted buffer-aware path. (e) Heat-map for data generation rate. (f) RP generation. (g) Path formation. (h) Adjusted buffer-aware path. (i) Heat-map for data generation rate. (j) RP generation. (k) Path formation. (l) Adjusted buffer-aware path.

path is further Adjusted Using the proposed Algorithm. These visual representations provide valuable insights into the variations in data generation, RP generation, path formation, and the adjusted paths based on buffer considerations. In fact, to conserve UAV's energy, we optimize its route by minimizing time spent at RPs and avoiding RPs that reduce the overall trajectory length. Both the reduced number of RPs and the optimized flight trajectory lead to significant preservation of UAV energy.

VII. Conclusion

The proposed approach tackles the central challenge of optimizing the UAV's path and service time at carefully selected RPs, leveraging buffer status and specific time windows as critical factors. By implementing the PCA-GLS method and employing the proposed innovative algorithm for dynamic service time adjustment based on buffer status, remarkable enhancements in the system's overall efficiency have been witnessed. Efficient planning the UAV's route and ensuring timely servicing of each RP within its designated time window have maximized data collection efficiency within the IoT network. The simulation results validates that our proposed strategy successfully optimized the UAV's trajectory, considerably reduced service time, and improved the efficiency of data collection. The proposed solution is suitable for several applications like, monitoring activities (forest, smart grids, border surveillance), disaster management etc.

However, in this work, the limited and short-lived energy sources of UAVs significantly constrain their operational duration. As a result, UAVs must frequently return to the base station (BS) to recharge their batteries, leading to substantial penalties in terms of mission delays and disruptions. Future work can address this issue by integrating wireless energy harvesting to recharge the onboard batteries through charging stations or vehicles.

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